

Eco-Friendly Chalk Making Using Eggshell Waste

Converting Kitchen Waste into Sustainable Products





The Problem & Our Solution

The Problem

Large amounts of eggshell waste are discarded daily from kitchens and food businesses — a missed opportunity for resource recovery.

Our Solution

Reuse eggshells as a raw material for making eco-friendly chalk — transforming waste into a useful product.

Our Aim

Create a low-cost, useful, and environmentally friendly chalk product accessible to schools and communities.

The Impact

Reduce food waste while developing a sustainable alternative to conventional chalk.

Eggshell Waste → Processing → Eco-Friendly Chalk



FIELD VISIT

Egg Maggie House

Our team visited Egg Maggie House to gather real-world insight into eggshell waste generation and to source raw materials for our chalk-making experiments.

→ Collect Waste Eggshells

Obtained eggshell samples for experimental use.

→ Study Disposal Methods

Learned how eggshells are currently discarded.

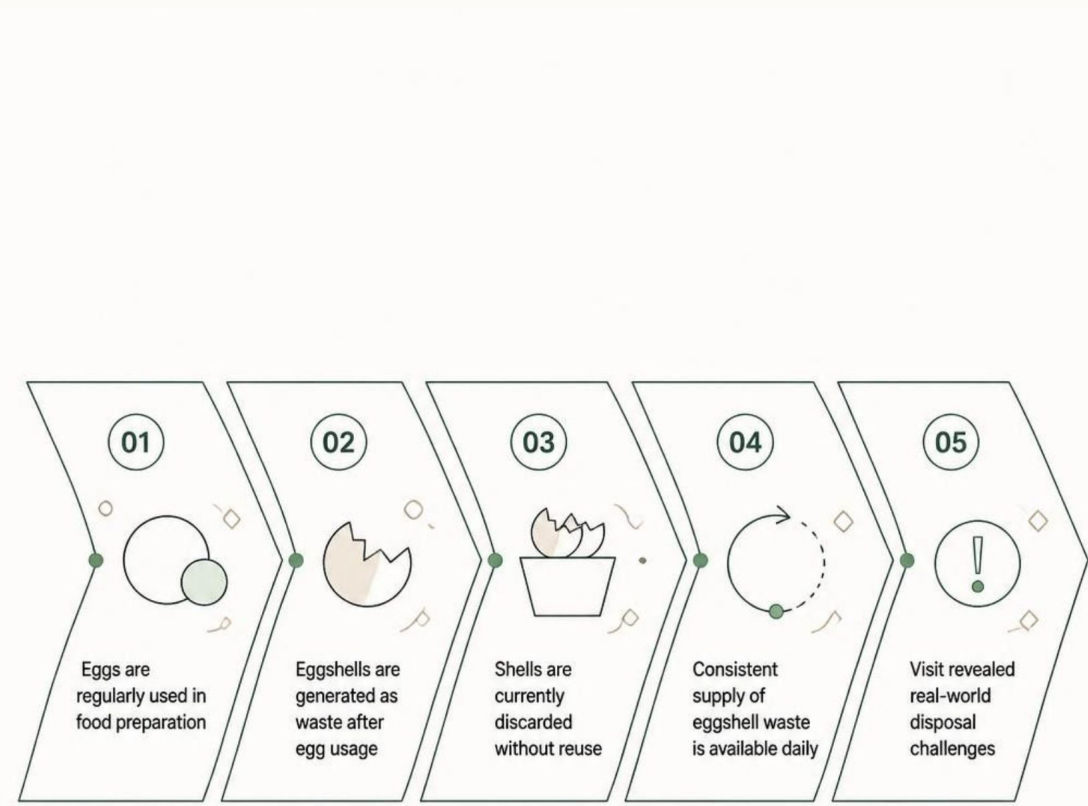
→ Understand Waste Volume

Observed the scale of daily eggshell waste generated.

→ Gather Project Data

Collected real-world data to inform our development process.

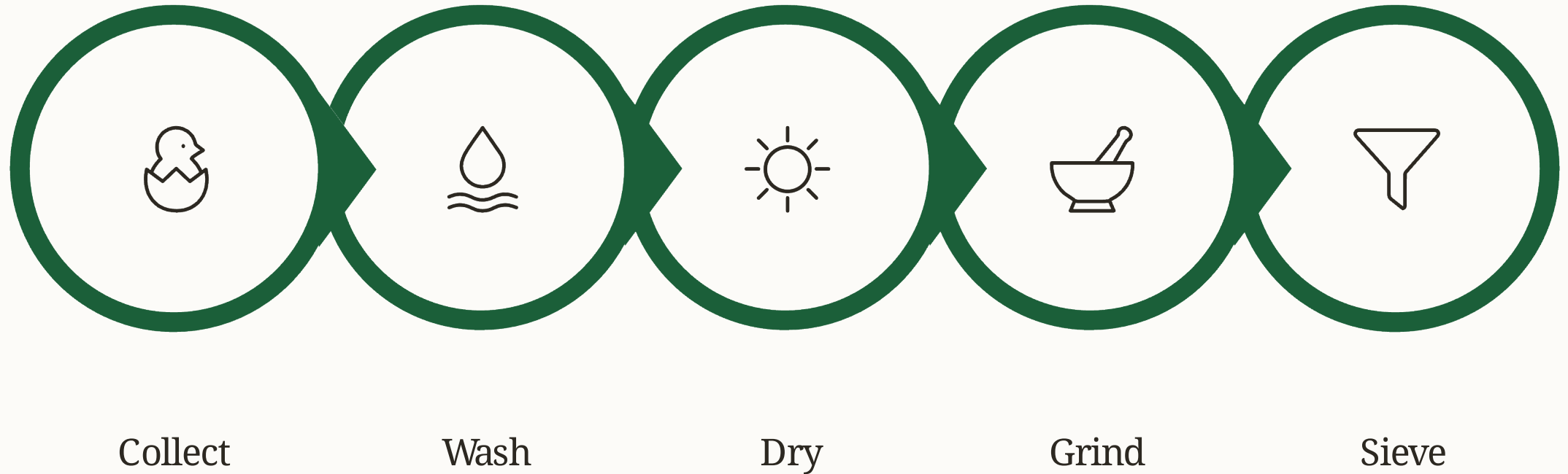
Information Collected During the Visit



Key Takeaways

- Eggs are regularly used in food preparation at the facility
- Eggshells are generated as waste after each use
- Shells are currently treated as waste and discarded
- Discussed the potential for collecting and reusing eggshell waste
- Confirmed availability of a consistent eggshell waste supply
- Collected eggshell samples representing daily operations

Process of Making Eco-Friendly Chalk



From raw waste to finished chalk — a fully reproducible, low-cost, eco-friendly process.



Results, Benefits & Conclusion

Results

- Eggshell waste was successfully processed as a raw material for chalk production
- Demonstrates a practical method to convert discarded waste into a useful product

Benefits

Reduces Waste

Diverts eggshells from landfills and kitchens

Low-Cost Alternative

Affordable substitute for conventional chalk

Conclusion

"Our project demonstrates how commonly discarded eggshell waste can be reused creatively to develop an eco-friendly chalk product, turning kitchen waste into a valuable educational resource."

Circular Economy

Promotes waste reuse and sustainable principles

Practical Learning

Real-world experience in sustainable product development